

Session 20

1.Create a table comparing Agile and Waterfall models across 5 factors relevant to software testing, such as requirements, feedback cycle, tester involvement, documentation, and flexibility.

Agile vs Waterfall – Comparison Table

Factor	Agile Model	Waterfall Model
Requirements	Requirements can change during development.	Requirements are usually fixed at the beginning.
Feedback Cycle	Frequent feedback is taken after every sprint.	Feedback is usually received after major phases or at the end.
Tester Involvement	Testers are involved from the beginning of each sprint.	Testing mainly starts after development is completed.
Documentation	Documentation is maintained as needed and updated regularly.	Detailed documentation is prepared before moving to the next phase.
Flexibility	Highly flexible; changes can be included in future sprints.	Less flexible; changes can be difficult and costly after development starts.

2.Pick any app you use daily (like Instagram, Zomato, or Paytm) and describe how the testing process would look different if the team used Agile instead of Waterfall. Mention at least 3 specific testing activities or practices that would change.

Testing Instagram Using Agile Instead of Waterfall

Suppose we are testing the **Instagram Post Upload** feature.

If Waterfall is used:

The requirements are finalized first, developers complete the feature, and then testers test the completed feature. Bugs are reported after development is largely finished.

If Agile is used:

The feature is divided into smaller user stories and tested continuously during short sprints.

3+ Testing Activities That Would Change

Testing Activity	Waterfall	Agile
Test Case Design	Test cases are prepared mainly before/after development is completed.	Testers design test cases while the feature is being developed.
Test Execution	Testing happens mainly after coding is finished.	Testing happens continuously during the sprint.
Bug Reporting	Bugs may be reported in a testing phase after development.	Bugs are reported immediately and discussed with developers.
Regression Testing	Usually performed after a development phase.	Regression testing is performed regularly after changes.
Tester–Developer Communication	More phase-based communication.	Daily communication through standups and sprint discussions.

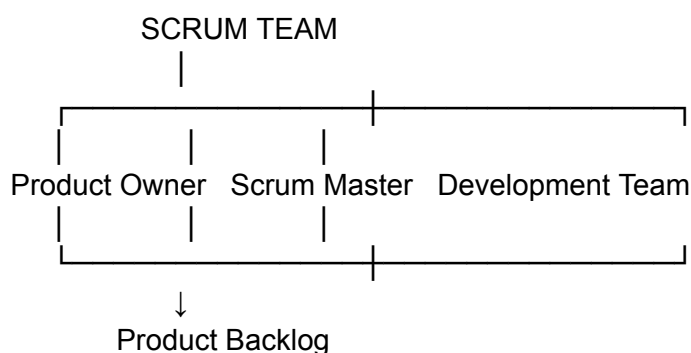
Example:

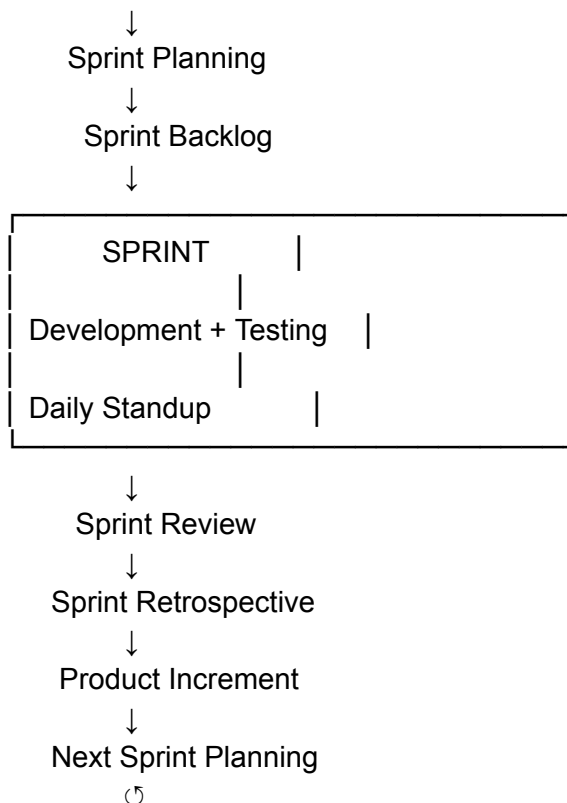
For Instagram's post upload, an Agile tester may test **image upload, caption, filters, posting, error messages, and network failure** within the same sprint rather than waiting until the entire application is completed.

3. Draw a simple workflow diagram showing the main roles (such as Scrum Master, Product Owner, Development Team, Tester) and ceremonies (like Sprint Planning, Daily Standup, Sprint Review) in a SCRUM process.

Simple SCRUM Workflow Diagram

You can draw this diagram in your notebook or assignment:





Main Roles

- **Product Owner:** Manages requirements and prioritizes the product backlog.
- **Scrum Master:** Helps the team follow Scrum practices and removes obstacles.
- **Development Team:** Develops and fixes the product.
- **Tester/QA:** Designs and executes tests, reports defects, and verifies fixes.

Main Ceremonies

- **Sprint Planning:** Team decides what work will be completed in the sprint.
- **Daily Standup:** Team discusses progress, today's work, and blockers.
- **Sprint Review:** Completed work is demonstrated to stakeholders.
- **Sprint Retrospective:** Team discusses what went well and what can be improved.

**4. For a food delivery app (like Swiggy), map out which testing activities should happen during each phase of a typical Agile sprint (e.g., backlog grooming, sprint planning, development, review).

Hint: Include activities like writing user stories, test case design, test execution, and bug reporting.**

Testing Activities During an Agile Sprint – Food Delivery App

Example: **Swiggy-like food delivery application**

Agile Phase	Testing Activities
1. Backlog Grooming / Refinement	Understand requirements, clarify user stories, identify acceptance criteria, identify possible test scenarios.
2. Sprint Planning	Select user stories for the sprint, estimate testing effort, identify testing requirements, plan test activities.
3. Development	Testers prepare test cases, review acceptance criteria, prepare test data, and discuss requirements with developers.
4. Testing / Execution	Execute functional, UI, integration, and negative test cases. Verify expected results.
5. Bug Reporting	Report defects with steps to reproduce, expected result, actual result, severity, and screenshots if required.
6. Bug Fix & Retesting	Developer fixes the defect. Tester retests the fix and performs regression testing on related functionality.
7. Sprint Review	Tester confirms that completed stories meet acceptance criteria and supports the demonstration of tested features.
8. Sprint Retrospective	Discuss testing problems, missed defects, process improvements, and ideas for the next sprint.

Example: Food Ordering User Story

User Story:

As a customer, I want to add food items to my cart so that I can place an order.

Testing activities:

1. Write acceptance criteria.
2. Design test cases for adding one or multiple items.
3. Test increasing/decreasing quantity.
4. Test removing an item.
5. Check cart total.
6. Test adding unavailable items.
7. Report bugs found.
8. Retest fixed bugs.
9. Perform regression testing.
10. Confirm the story is ready for Sprint Review.

Simple Sprint Flow

Backlog Grooming



Sprint Planning



User Stories + Acceptance Criteria



Development



Test Case Design



Test Execution



Bug Reporting



Bug Fix



Retesting + Regression Testing



Sprint Review



Sprint Retrospective



Next Sprint

Conclusion: In Agile, testing is a **continuous activity throughout the sprint**, rather than being treated as a separate phase after development.